

Tuple creation

```
In [1]: t1=()          # empty tuple

In [2]: t2=(1,2,3,4)   # int no.

In [3]: t3=(3.4, 5.2,6.6)   # float no.

In [4]: t4= ('Pragati','Tejal')   # string

In [5]: t5=( True,False)    # bool

In [7]: t6=(1,3.3,True,'Pragati',(1+4j))    # mix datatypes

In [10]: t7=(1,(2,3.4),{'name':'Pragati','age': 24},[2,3,[5,6]])    # nested tuple

In [11]: len(t6)

Out[11]: 5
```

tuple indexing

```
In [13]: t4[1]

Out[13]: 'Tejal'

In [19]: t4[1][4]

Out[19]: 'l'

In [20]: t5[1]

Out[20]: False

In [21]: t5[0][3]          # in bool, index in index not allow
```

```
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TypeError                                Traceback (most recent call last)
Cell In[21], line 1
----> 1 t5[0][3]

TypeError: 'bool' object is not subscriptable
```

```
In [22]: t5[-1]

Out[22]: False
```

tuple slicing

```
In [23]: tuple=('Pragati','Tejal','You','me','one','two')
```

```
In [24]: tuple[3:6]
```

```
Out[24]: ('me', 'one', 'two')
```

```
In [25]: tuple[0:6]
```

```
Out[25]: ('Pragati', 'Tejal', 'You', 'me', 'one', 'two')
```

```
In [26]: tuple[:]
```

```
Out[26]: ('Pragati', 'Tejal', 'You', 'me', 'one', 'two')
```

```
In [27]: tuple[-3:]
```

```
Out[27]: ('me', 'one', 'two')
```

```
In [28]: tuple[3:-1]      # last exclude
```

```
Out[28]: ('me', 'one')
```

```
In [29]: tuple[3:]
```

```
Out[29]: ('me', 'one', 'two')
```

```
In [30]: tuple[-5:-2]
```

```
Out[30]: ('Tejal', 'You', 'me')
```

change & remove elem

```
In [31]: tuple=('Pragati','Tejal','You','me','one','two')
```

```
In [32]: tuple[2]= 'Three'      # immutable (can't change)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[32], line 1  
----> 1 tuple[2]= 'Three'  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [33]: del tuple[3]      # can't delete specific elem
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[33], line 1  
----> 1 del tuple[3]  
  
TypeError: 'tuple' object doesn't support item deletion
```

```
In [34]: del tuple      # delete entire tuple
```

```
In [35]: tuple
```

```
Out[35]: tuple
```

loop through a tuple

```
In [36]: tuple=('Pragati','Tejal','You','me','one','two')
```

```
In [37]: for i in tuple:
          print(i)
```

```
Pragati
Tejal
You
me
one
two
```

```
In [38]: tuple=('Pragati','Tejal','You','me','one','two')
```

```
In [40]: for i in enumerate (tuple):
          print(i)
```

```
(0, 'Pragati')
(1, 'Tejal')
(2, 'You')
(3, 'me')
(4, 'one')
(5, 'two')
```

tuple membership

```
In [41]: tuple=('Pragati','Tejal','You','me','one','two')
```

```
In [43]: 'one' in tuple
```

```
Out[43]: True
```

```
In [44]: 'tejal' in tuple    # Tejal
```

```
Out[44]: False
```

```
In [45]: if 'You' in tuple:
          print('two is present in tuple')
        else:
          print('two is not present in tuple')
```

```
two is present in tuple
```

```
In [46]: if 'tejal' in tuple:
          print('tejal is present in tuple')
        else:
          print('tejal is not present in tuple')
```

```
tejal is not present in tuple
```

index position

```
In [47]: tuple=('Pragati','Tejal','You','me','one','two')
```

```
In [50]: tuple.index('one')
```

```
Out[50]: 4
```

```
In [51]: tuple.index(two)    # two is string
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[51], line 1  
----> 1 tuple.index(two)    # two is string  
NameError: name 'two' is not defined
```

```
In [52]: tuple.count('me')
```

```
Out[52]: 1
```

sorting

```
In [53]: tuple=(2,4.4,6,9,30,True)
```

```
In [56]: sorted(tuple)
```

```
Out[56]: [True, 2, 4.4, 6, 9, 30]
```

```
In [57]: sorted(tuple,reverse=True)
```

```
Out[57]: [30, 9, 6, 4.4, 2, True]
```

```
In [ ]:
```